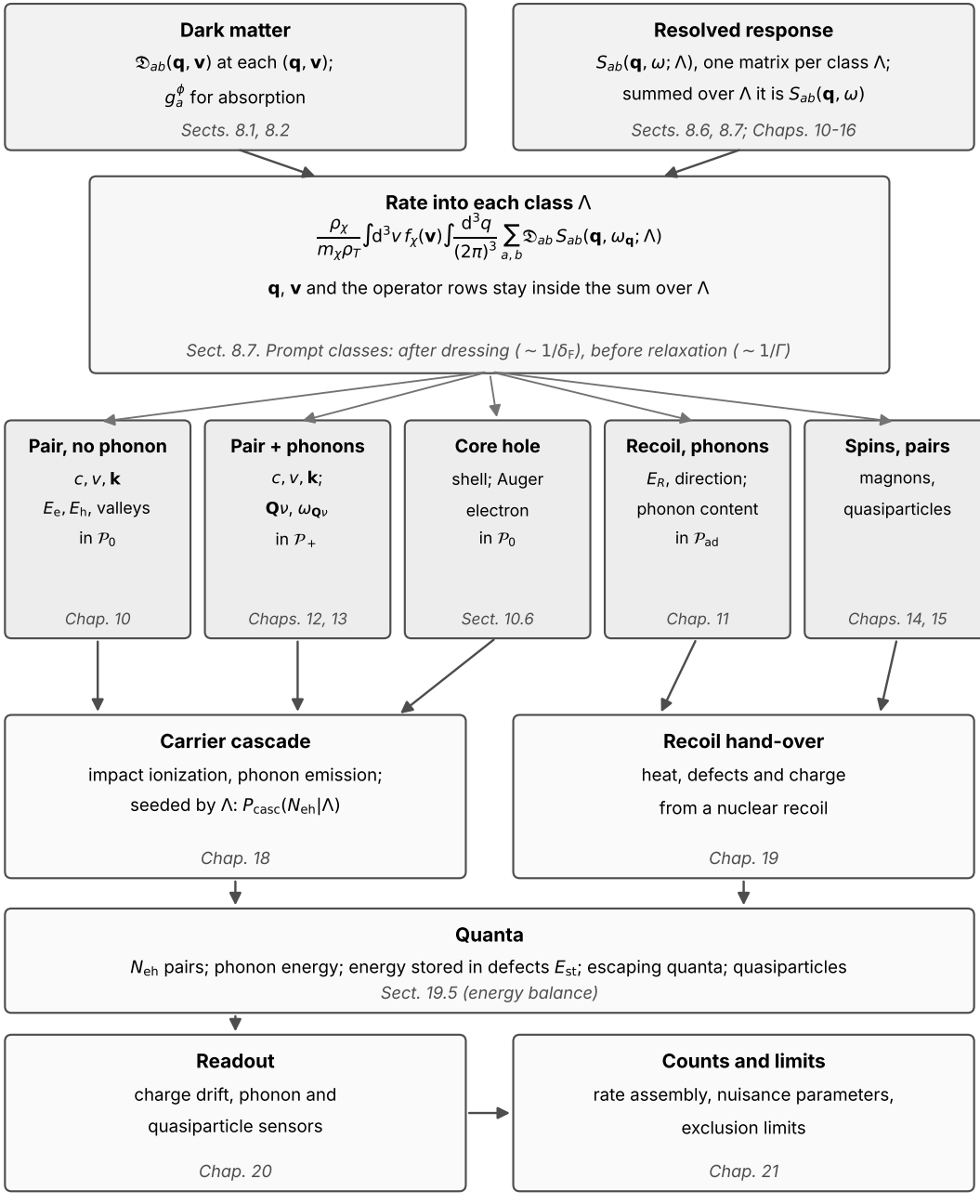

Dark matter
$\mathfrak{D}_{ab}(\mathbf{q}, \mathbf{v})$ at each $(\mathbf{q}, \mathbf{v})$;
g_a^ϕ for absorption
Sects. 8.1, 8.2
Resolved response
$S_{ab}(\mathbf{q}, \omega; \Lambda)$, one matrix per class Λ;
summed over Λ it is $S_{ab}(\mathbf{q}, \omega)$
Sects. 8.6, 8.7; Chaps. 10-16
Rate into each class Λ
$\frac{\rho_\chi}{m_\chi \rho_T} \int \mathrm{d}^3 v\, f_\chi(\mathbf{v}) \int \frac{\mathrm{d}^3 q}{(2\pi)^3} \sum_{a,b} \mathfrak{D}_{ab} S_{ab}(\mathbf{q}, \omega_{\mathbf{q}}; \Lambda)$
$\mathbf{q}$, $\mathbf{v}$ and the operator rows stay inside the sum over Λ
Sect. 8.7. Prompt classes: after dressing ($\sim 1/\delta_F$), before relaxation ($\sim 1/\Gamma$)
Pair, no phonon
$c, v, \mathbf{k}$
E_e, E_h, valleys
in $\mathcal{P}_0$
Chap. 10
Pair + phonons
$c, v, \mathbf{k}$;
$\mathbf{Q}\nu$, $\omega_{\mathbf{Q}\nu}$
in $\mathcal{P}_+$
Chaps. 12, 13
Core hole
shell; Auger
electron
in $\mathcal{P}_0$
Sect. 10.6
Recoil, phonons
E_R, direction;
phonon content
in $\mathcal{P}_{ad}$
Chap. 11
Spins, pairs
magnons,
quasiparticles
Chaps. 14, 15
Carrier cascade
impact ionization, phonon emission;
seeded by Λ: $P_{casc}(N_{eh}|\Lambda)$
Chap. 18
Recoil hand-over
heat, defects and charge
from a nuclear recoil
Chap. 19
Quanta
N_{eh} pairs; phonon energy; energy stored in defects E_{st}; escaping quanta; quasiparticles
Sect. 19.5 (energy balance)
Readout
charge drift, phonon and
quasiparticle sensors
Chap. 20
Counts and limits
rate assembly, nuisance parameters,
exclusion limits
Chap. 21